# YiRage (YPK) — Multi-backend LLM Megakernel Compiler & Runtime

github.com/chenxingqiang/YiRage · based on Mirage (CMU) + multi-backend / RuntimeFusion extensions

## 1. Serving / embedding

**vLLM**
keeps schedule + KV
RF selects capsules

**SGLang**
decode-loop-erdecode
same FusionPlan cache

**Python API**
PersistentKernel
HardwareRegistry

**UnifiedCompiler**
graph + backend pick
Cython CoreBridge

## 2. RuntimeFusion

**FusionPlan cache**
search / profile results
per backend + shape

**FusionCapsule**
one decode-step kernel
selected each token

**Device memory**
persistent workspace
KV-friendly layout

cache

## 3. Compiler / search

**Kernel graph IR**
matmul / attn / norm
mark inputs / outputs

**Superoptimize**
hardware-aware search
15 candidate dims

**Fingerprint**
numeric verify
before accept

**Autotune**
grid / tile / pipeline
occupancy model

## 4. Threadblock operators + transpilers

**MatMul**
Tensor Core / Cube

**Attention / GQA**
Flash-style tiles

**RMSNorm / SwiGLU**
fused decode path

**Triton codegen**

**NKI / MLIR**

**CUDA / HIP JIT**

## 5. Backend registry (12 targets; factory + thread-safe singleton)

**CUDA**
Tensor Core
bank conflict

**ROCm**
HIP / MFMA

**MPS**
Apple Metal
unified mem

**Ascend**
AI Core / Cube
BiSheng

**MACA**
MetaX

**TPU / XPU**
NKI / Intel

**MLIR / CPU**
LLVM / SIMD

**FPGA**
experimental

Serving keeps scheduling/KV; compiler searches fused kernels; RuntimeFusion injects FusionCapsules into vLLM/SGLang decode.